

PROFESSIONAL SUMMARY

Robotics researcher and developer, with a PhD in Computer science, specializing in heuristic multi-robot task allocation, prioritized exploration, and vision-based algorithms for real-time localization and manipulation. Demonstrated expertise in developing stereo vision and SLAM techniques alongside efficient scheduling and path-planning algorithms, validated through impactful industry and academic roles at Mujin Corp., ABB Inc., and UNC Charlotte.

EXPERIENCE

- Robotic Software Developer** *Mujin Corp., Suwanee, GA* 2024 — Current
 - Developed heuristic scheduling algorithms for warehouse execution system for automatic case picking warehouse.
 - Developed storage assignment algorithms to reduce pallet movement by over 30%.
 - Developed pytest environments to simulate mobile robot warehouse movement with integrated failure testing.
 - Designed warehouse layout improvements and AGV pathways for ~40% improvement.
 - Gained significant simulation, real-world testing, deployment, and production experience overseeing automated warehouse system for 7 months.
- Graduate Research Assistant** *UNC Charlotte* 2017 — 2024
 - Developed 3D simulation environments in Blender and Gazebo for multi-robot exploration.
 - Developed heuristic path planning algorithms for single and multi-robot exploration in time-sensitive environments.
 - Implemented multi-robot waypoint navigation for quadcopters in outdoor simulation environments using Robot Operating System (ROS) and AirSIM as a part of DARPA's OFFensive Swarm-Enabled Tactics (OFFSET) program.
 - Optimized low light collision scene reconstruction by identifying drone trajectories and light positions. Collaborated with NC-DOT and RTI to reduce data collection time by 95%.
 - Conducted field experiments with UAV towards human tracking in cluttered environments.
 - Organized computational workflows and data pipelines for multiple data sources for path computation and navigation.
 - Quick prototyping of proof-of-concept code and executing code in C++ and Python in tiny computer platforms.
 - Mentored and managed several undergraduate students. Notable ones were to create an AirSIM simulation environment of a university building, and real-time point clouds using multiple stereo cameras.
 - Identified potential impacts of an autonomous shuttle in university in terms of usability and ridership. Collected Lidar and multiple stereo camera data from autonomous shuttle using ROS2 totaling 67TB over 7 weeks.
- Research Intern** *ABB Inc., Raleigh, NC* 2022
 - Implemented multi view stereo for 6D tracking of objects in tote boxes for real time pick and place using robot manipulator
- Research Assistant** *Robotics Research Center, IIIT Hyderabad* 2015 — 2017
 - Implemented monocular and stereo visual SLAM with loop detection in Matlab and C++ simulations and experiments on outdoor mobile robots
 - Created a performance evaluation platform by creating ground truth data using April tags
 - Developed algorithms for single-robot and multi-robot pose graph SLAM even in presence of 60% outliers
- Instructor of Record** *UNC Charlotte, NC* 2022
 - Prepared course materials, assignments, and taught ITCS 4150 Mobile Robotics in Fall 2022
 - Taught Python programming, Linux, ROS communication, Turtlebot3 packages, localization and mapping
- Graduate Teaching Assistant** *UNC Charlotte, NC* 2017 — 2023
 - Created video tutorials on Python programming, Linux Virtual Machine on Windows and Mac, and ROS
 - Created and graded programming assignments in on Robot Operating System (ROS), Python programming, robot kinematics, inverse Kinematics, odometry, localization and mapping in Mobile Robotics and Intelligent Robotics courses in a class of over 40 students for 6 semesters.
 - Developed scripts to assist grading reducing per student grading time by 60%.
- Summer Intern — Open source developer** *Google Summer of Code* 2013
 - Contributed a set of local image editing tools in photo editing tool, digiKam, using C++ and Qt framework.
- Summer Intern — Open source developer** *Season of KDE* 2012
 - Developed automatic noise estimation and reduction in images in digiKam

EDUCATION

The University of North Carolina at Charlotte

Charlotte, NC

Ph.D. Candidate in Computer Science — GPA 4.0

2017 – 2024

Dissertation: Prioritized robotic exploration with dynamic deadlines

International Institute of Information Technology Hyderabad

Hyderabad, India

MS by Research in Computer Science — GPA 8.14/10.0

2014 – 2017

Thesis: Applications of spectral graph theory in Pose-graph SLAM

National Institute of Technology Durgapur

Durgapur, India

Bachelor of Technology (B. Tech.) in Computer Science — GPA 8.37/10.0

2010 – 2014

Thesis: Creating a quadcopter and crowd-sourced data analysis using a GPS logger

SELECTED PROJECTS

- Developed semantic information-based greedy path planning algorithm for robot exploration under deadlines resulting in performance improvement in percent explored over 10%. (2021)
- Developed several heuristic algorithms for single robot and multi-robot for robot exploration. (2022 — 2024)
- Semantic perception using a single-scan Lidar data to classify the building structures using Python3 and ROS. (2020)
- Created a reinforcement learning environment in OpenAI Gym for robot exploration. (2020)
- Path planning for a Fetch manipulator to reach objects on table using reinforcement learning. (2020)
- Reconstruction Quality Evaluation using image metrics and usability testing using rank correlation, inter-coder reliability metric. Project resulted in 95% decrease in time taken to capture data in collision scenes at night. (2020)
- Created a quadcopter using brushless DC motors, speed controller, flight controller board, and a light wood frame as a part of undergraduate thesis. PID calibrated motors and frame size for stable flight. (2014)
- Developed a Image editor with GUI support using C++, OpenCV and Qt QML. (2014)
- Auto-crop of stitched images to create clean panorama image. Reduced time to crop stitched images by 98%. (2013)

PUBLICATIONS

- *Prioritized Robotic Exploration with Deadlines: A Comparison of Greedy, Orienteering, and Profitable Tour Approaches* IEEE International Conference of Robotics and Automatics (ICRA) 2023, **Sayantana Datta**, Srinivas Akella
- *Prioritized Indoor Exploration with a Dynamic Deadline*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2021, **Sayantana Datta**, Srinivas Akella
- *Low-light collision scene reconstruction using unmanned aerial systems*, 2019, Joe Eyerman, Bradley Mooring, Maureen Catlow, **Sayantana Datta**, Srinivas Akella
- *Multi-trajectory Pose Correspondences using Scale-dependent Topological Analysis of Pose-graphs*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2017, **Sayantana Datta**, Avinash Sharma, K. Madhava Krishna
- *Pose-graph Robustification via Heat-Kernel Analysis*, IEEE Conference on Decision and Control (CDC), 2016, **Sayantana Datta**, Siddharth Tourani, Avinash Sharma, K. Madhava Krishna

TECHNICAL SKILLS

Languages: Python, C/C++, R, Java, Matlab

Frameworks and Tools: Robot Operating System(ROS), Gazebo Simulator, OpenRAVE, AirSim, MoveIt, Git, Docker, SVN

Libraries: NumPy, Scipy, Pytest, Matplotlib, Seaborn, Pandas, OpenCV, OpenGL, Open3D, TensorFlow, CUDA, PyTorch, Keras, MRPT, OMPL, g2o, boost

Robots worked with: Turtlebot 1,2, and 3, AgileX Limo, Clearpath Husky and Jackal, UR5e manipulator, Pioneer 3DX Quicktron M100, M150

AWARDS AND VOLUNTARY SERVICE

Awards

- Graduate Life Fellow (2021 — 2022) for improving interdisciplinary graduate student connections
- Graduate School Summer Fellowship (2023) for summer research
- Graduate Assistant Support Plan (GASP) (2017—2023) at UNC Charlotte during doctoral studies

Reviewer

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Automation Science and Engineering (CASE)

EXTRA CURRICULARS

2022 — 2023	Mentor at Mentor Collective, where I mentored four graduate students
2021 — 2022	Graduate Life Fellow at Graduate School, UNC Charlotte
2010 — 2014	Member of Linux Users Group to promote open source software and usage